

8 Factor Analysis: Practical Methods

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Factor Analysis: Practical Models

Recall from last chapter that the classical factor analysis model is given by

$$X = \mu + LF + \varepsilon$$

where:

- X is a random vector of p observed variables (indicators).
- μ is a constant vector of dimension p .
- L is a $p \times m$ constant matrix known as the matrix of *factor loadings*.
- $F \sim \mathcal{N}(0, I_m)$ is a vector of m latent factors (unobserved variables) that are assumed to be normally distributed with mean zero and identity covariance matrix.
- $\varepsilon \sim \mathcal{N}(0, \Psi)$ is a vector of unique errors (specific variances) that are also assumed to be normally distributed with mean zero and diagonal covariance matrix $\Psi = \text{diag}(\sigma_1^2, \dots, \sigma_p^2)$.
- The factors F and the errors ε are assumed to be independent, in particular

Assesing Factorability

The first step in applying factor analysis to a dataset is to assess whether the data is suitable for factor analysis, i.e., whether the observed variables exhibit sufficient correlations to justify the use of a factor model. Several statistical tests and measures can be used to evaluate the factorability of a dataset:

1. **Correlation Matrix Examination:** A preliminary step is to visually inspect the correlation matrix of the observed variables. Look for patterns of high correlations (e.g., > 0.3) among variables, which suggest that they may share common underlying factors.
2. **Bartlett's Test of Sphericity:** This test evaluates the hypothesis that the correlation matrix is an identity matrix, which would indicate that the variables are uncorrelated and unsuitable for factor analysis. A significant result ($p\text{-value} < 0.05$) suggests that the variables are correlated enough to proceed with factor analysis.
3. **Kaiser-Meyer-Olkin (KMO) Measure of Sampling Adequacy:** The KMO statistic measures the proportion of variance in the observed variables that can be

Deciding the Number of Factors

Determining the appropriate number of factors to retain in a factor analysis is a crucial step that can significantly impact the results and interpretations. Several methods can be employed to decide on the number of factors:

1. **Eigenvalue Criterion (Kaiser Criterion):** Retain factors with singular values greater than 1. This method is based on the idea that a factor should explain at least as much variance as a single observed variable.
2. **Scree Plot:** A scree plot displays the singular values in descending order. Look for an "elbow" point where the slope of the curve changes dramatically. The number of factors to retain is typically determined by the number of points before this elbow.
3. **Parallel Analysis:** This method involves comparing the observed singular values with those obtained from random data of the same size. Factors are retained if their singular values exceed those from the random data.
4. **Model Fit Indices:** For more advanced factor analysis methods (e.g., confirmatory

Extracting Factors

Once the number of factors has been determined, the next step is to extract the factors from the data. Several methods are available for factor extraction:

1. **Principal Axis Factoring (PAF):** This method focuses on the shared variance among variables and is suitable when the goal is to identify underlying constructs. It iteratively estimates communalities and extracts factors based on the reduced correlation matrix.
2. **Maximum Likelihood (ML) Factor Analysis:** This method estimates factor loadings by maximizing the likelihood of the observed data given the model. It allows for statistical testing of hypotheses about the factor structure and provides goodness-of-fit indices.

Rotation of Factors

After extracting factors, it is often beneficial to rotate them to achieve a simpler and more interpretable structure. Rotation can be either orthogonal (factors remain uncorrelated) or oblique (factors are allowed to correlate). Common rotation methods include:

1. **Varimax Rotation:** An orthogonal rotation method that maximizes the variance of squared loadings within each factor, leading to a simpler structure where each variable loads highly on one factor and low on others.
2. **Quartimax Rotation:** Another orthogonal rotation method that simplifies the rows of the loading matrix, aiming to have each variable load highly on as few factors as possible.
3. **Equamax Rotation:** A hybrid orthogonal rotation method that combines the goals of varimax and quartimax, balancing the simplification of both rows and columns of the loading matrix.

Computing Factor Scores

Once the factor loadings have been estimated and the factors have been rotated (if applicable), the next step is to compute factor scores for each observation in the dataset. Factor scores represent the estimated values of the latent factors for each individual based on their observed variable scores. Several methods can be used to compute factor scores:

1. **Regression Method:** This method estimates factor scores by regressing the observed variables on the estimated factor loadings. It provides unbiased estimates of factor scores but may not preserve the original variance of the factors.
2. **Bartlett Method:** This method computes factor scores by minimizing the residual variance of the observed variables. It tends to produce factor scores that are more reliable and have lower measurement error.
3. **Anderson-Rubin Method:** This method generates factor scores that are uncorrelated and standardized. It is particularly useful when the factors are assumed to be orthogonal.

Assesing Model Fit

After fitting a factor analysis model, it is important to assess how well the model fits the data. Several statistical tests and fit indices can be used to evaluate model fit:

1. **Chi-Square Test of Model Fit:** This test evaluates the null hypothesis that the model-implied covariance matrix matches the observed covariance matrix. A non-significant result ($p\text{-value} > 0.05$) suggests that the model fits the data well.
2. **Root Mean Square Error of Approximation (RMSEA):** This index measures the discrepancy between the observed and model-implied covariance matrices per degree of freedom. Values less than 0.06 indicate a good fit.
3. **Comparative Fit Index (CFI) and Tucker-Lewis Index (TLI):** These indices compare the fit of the specified model to a null model. Values greater than 0.95 indicate a good fit.
4. **Standardized Root Mean Square Residual (SRMR):** This index measures the average discrepancy between observed and predicted correlations. Values less than 0.08 indicate a good fit.

Visualizing the Model

Visualizing the results of a factor analysis can help in interpreting the factors and understanding the relationships among variables. Common visualization techniques include:

1. **Scree Plot:** A scree plot displays the singular values in descending order, helping to visualize the number of factors to retain.
2. **Factor Loading Plot:** A scatter plot of factor loadings can help visualize how variables load on different factors, especially after rotation.
3. **Biplot:** A biplot combines factor scores and loadings in a single plot, allowing for the visualization of both observations and variable relationships in the factor space.
4. **Heatmap of Loadings:** A heatmap can be used to visualize the magnitude of factor loadings across variables and factors, highlighting patterns and clusters.

Remark

Notice that Factor Loading Plots and Biplots only make sense when we have 2 or 3

Example: Factor Analysis Biplot of the Holzinger Swineford 1939 Dataset

```
data(HolzingerSwineford1939)

# Select the relevant tests
hs_data <- HolzingerSwineford1939[, c("x1", "x2", "x3", "x4", "x5", "x6", "x7", "x8", "x9", "x10", "x11", "x12", "x13", "x14", "x15", "x16", "x17", "x18", "x19", "x20", "x21", "x22", "x23", "x24", "x25", "x26", "x27", "x28", "x29", "x30", "x31", "x32", "x33", "x34", "x35", "x36", "x37", "x38", "x39", "x40", "x41", "x42", "x43", "x44", "x45", "x46", "x47", "x48", "x49", "x50", "x51", "x52", "x53", "x54", "x55", "x56", "x57", "x58", "x59", "x60", "x61", "x62", "x63", "x64", "x65", "x66", "x67", "x68", "x69", "x70", "x71", "x72", "x73", "x74", "x75", "x76", "x77", "x78", "x79", "x80", "x81", "x82", "x83", "x84", "x85", "x86", "x87", "x88", "x89", "x90", "x91", "x92", "x93", "x94", "x95", "x96", "x97", "x98", "x99", "x100")]

# Fit 2-factor FA with varimax rotation
fa_res <- factanal(hs_data, factors = 2, rotation = "varimax", scores = "re")

# Extract factor loadings for biplot
loadings_matrix <- as.matrix(fa_res$loadings)
loadings_df <- data.frame(
  Variable = rownames(loadings_matrix),
  Factor1 = loadings_matrix[, 1],
  Factor2 = loadings_matrix[, 2]
)
```